

LUXEUIL AB, France

WMO: 072920

Lat: 47.787N

Lon: 6.364E

Elev: 913

StdP: 14.22

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.3	(c) 20.1	(d) 9.8	(e) 9.4	(f) 18.8	(g) 15.4	(h) 12.4	(i) 23.7	(j) 25.8	(k) 47.0	(l) 23.2	(m) 45.7	(n) 2.7	(o) 60	(p) 0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	88.3	69.1	84.9	68.0	81.8	66.9	71.2	83.7	69.7	81.5	68.2	79.1	5.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	104.1	75.9	65.7	98.2	74.1	64.3	93.3	72.5	35.6	83.9	34.2	81.7	32.9	79.3	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.2	(b) 16.7	(c) 13.9	(e) 8.3	(f) 93.1	(g) 6.1	(h) 3.9	(i) 3.9	(j) 95.9	(k) 0.3	(l) 98.2	(m) -3.2	(n) 100.4	(o) -7.6	(p) 103.2		
(a) 20.2	(b) 16.7	(c) 13.9	(e) 7.9	(f) 74.7	(g) 5.8	(h) 1.7	(i) 3.7	(j) 75.9	(k) 0.3	(l) 76.9	(m) -3.0	(n) 77.8	(o) -7.3	(p) 79.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	35.7	37.7	43.8	49.8	56.9	63.4	66.5	65.9	58.6	52.2	42.6	36.8		
	DBStd	12.85	7.56	7.26	6.61	6.51	6.30	6.77	5.67	6.04	5.83	6.70	7.11	7.62		
	HDD50	1798	443	344	209	83	13	1	0	0	3	56	237	409		
	HDD65	5413	907	763	656	456	260	110	49	63	204	399	673	873		
	CDD50	2132	1	1	18	77	228	403	512	492	263	123	13	1		
	CDD65	274	0	0	0	0	10	63	96	90	14	1	0	0		
	CDH74	3408	0	0	0	34	194	793	1172	1019	181	15	0	0		
	CDH80	1056	0	0	0	1	27	243	406	348	31	0	0	0		
(14) Wind	WSAvg	5.3	5.9	6.1	6.1	5.6	5.2	4.9	4.8	4.5	4.5	4.8	5.3	5.8		
(15) Precipitation	PrecAvg	47.9	4.1	3.8	3.4	3.2	4.1	3.7	3.7	3.7	3.8	4.5	4.8	5.0		
	PrecMax	66.5	8.5	7.5	10.8	7.4	6.5	7.4	8.0	8.2	8.0	8.9	10.7	10.3		
	PrecMin	35.2	1.0	0.9	0.4	0.6	1.0	1.0	1.1	0.3	0.9	0.6	0.9	0.5		
	PrecStd	6.7	2.1	2.0	2.4	1.9	1.3	1.7	1.6	1.9	2.0	2.2	2.6	2.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.9	61.0	69.6	78.3	84.2	91.4	93.6	92.9	84.1	76.2	64.2	56.4		
		MCWB	50.0	49.2	53.3	59.3	66.5	70.0	69.6	70.8	66.2	62.7	55.4	51.0		
	2%	DB	52.2	56.1	65.5	73.2	79.3	86.2	88.2	87.6	79.5	71.2	58.9	52.9		
		MCWB	48.7	47.6	52.2	57.3	63.7	68.5	68.7	69.3	66.4	61.6	53.8	49.5		
	5%	DB	49.7	52.2	60.9	68.6	75.7	82.5	84.6	83.8	75.3	66.9	55.8	50.4		
		MCWB	46.9	46.4	50.0	55.2	61.5	67.4	67.7	68.2	64.3	59.6	52.0	47.9		
	10%	DB	46.6	48.9	56.4	64.4	71.8	78.7	80.9	80.0	71.4	62.9	52.8	47.9		
		MCWB	44.5	44.9	48.2	52.9	59.6	65.8	66.4	67.0	61.8	57.8	49.7	45.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.0	51.3	55.7	61.2	68.1	73.0	72.8	73.2	71.4	65.1	56.9	52.3		
		MCDB	53.6	56.4	65.6	75.2	80.4	85.8	85.9	86.4	77.7	72.2	61.1	54.3		
	2%	WB	49.3	49.1	53.2	58.3	65.4	70.4	70.7	71.1	67.3	62.6	54.7	50.1		
		MCDB	51.6	54.4	61.4	69.8	76.8	82.4	83.8	83.7	75.8	69.3	58.0	52.1		
	5%	WB	46.9	47.2	51.3	56.2	62.9	68.4	69.1	69.4	65.2	60.4	52.2	48.2		
		MCDB	49.3	51.2	58.6	66.6	72.6	79.8	81.5	80.8	73.5	65.6	55.3	50.4		
	10%	WB	44.6	45.2	49.4	54.0	60.6	66.6	67.4	67.8	62.9	58.2	50.1	45.9		
		MCDB	46.5	48.7	55.5	62.8	69.4	77.1	78.2	78.2	69.8	62.6	52.6	47.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	15.0	19.2	21.8	21.3	22.1	22.0	22.1	20.8	18.2	13.2	11.3		
		MCDBR	14.3	23.0	29.7	31.1	29.3	28.4	29.7	29.6	27.6	23.9	18.3	14.1		
		MCWBR	11.0	15.2	17.5	16.8	14.9	13.1	13.0	13.5	15.4	14.8	13.0	10.9		
	5% WB	MCDBR	12.8	18.6	24.4	27.6	25.7	25.4	26.5	26.5	23.8	21.7	16.6	13.0		
		MCWBR	10.4	13.1	15.0	15.2	13.7	12.4	12.6	12.8	14.1	14.1	12.4	10.5		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.333	0.362	0.396	0.403	0.409	0.405	0.401	0.377	0.361	0.329	0.307		
	taud		2.446	2.390	2.330	2.252	2.273	2.291	2.298	2.325	2.371	2.405	2.446	2.445		
	Ebn at Noon		244	261	271	271	273	271	271	266	262	247	233	230		
	Edh at Noon		21	27	34	40	41	40	40	37	32	27	21	19		
(44) All-Sky Solar Radiation	RadAvg		325	577	970	1374	1608	1880	1782	1547	1177	724	376	262		
	RadStd		46	95	148	200	167	169	163	149	113	63	43	50		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (79 neighbors)	N/A	N/A	N/A	N/A	+0.67	+0.47	N/A	N/A	N/A	N/A

Nomenclature: See separate page